

Theory of Mechanism and Machine

(Week-6)

Acceleration in Mechanism:

- *Acceleration diagram.* It is the graphical representation of the accelerations of the various links of a mechanism drawn on a suitable scale.
- The determination of acceleration of various links is important from the point of view of calculating the forces and torques in the various links to carry out the dynamic analysis.
- As acceleration is proportional to the square of the speed, i.e., $acceleration = \omega^2 \times r$, hence if the speed of the machines becomes two times, the centripetal force will become four times. Thus the acceleration diagrams are, therefore, fundamental to stress analysis of mechanism.

Acceleration Diagram for a Link:

The acceleration of any point in a mechanism can be determined by two important methods, which are :

(i) Analytical method and

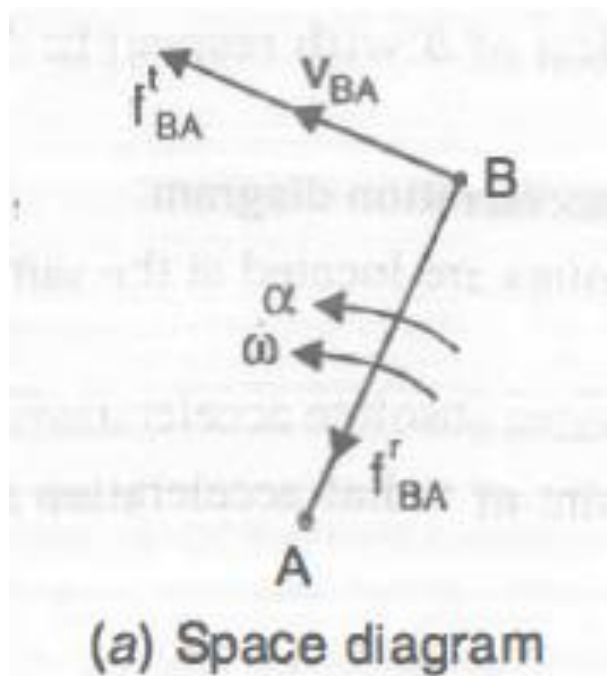
(ii) Graphical or acceleration diagram method.

- If an expression for displacement in terms of time (t) is known, then analytical method can be applied. But for most of mechanism, the expression for displacement cannot be determined easily. Hence graphical or acceleration diagram method is generally used.

Acceleration Diagram Method:

The Fig. (a) shows a rigid link AB. Let the point B is moving with respect to point A. To find the acceleration of B relative to A, find acceleration of B assuming point A to be fixed and if A is assumed fixed, then only possible motion of B will be rotation about A as centre.

- Let ω and α are angular velocity and acceleration of link AB respectively.



As point B is rotating about A, the velocity of B is changing in magnitude and direction. Hence the acceleration of the point B will have two components:

1) Radial component (centripetal component) which is due to angular velocity. It will be acting along BA (or parallel to BA) and will be directed from B towards A. The magnitude of this component will be v^2/r or $\omega^2 \times r$. Hence radial component of B with respect to A:

$$f^r_{BA} = \frac{V_{BA}^2}{BA} \quad \text{or} \quad \omega^2 \times BA$$

2) Tangential component which is due to angular acceleration (α). This acts parallel to the velocity or it is perpendicular to AB. The magnitude of this component is $\alpha \times BA$. Hence tangential component of B with respect to A, is given by-

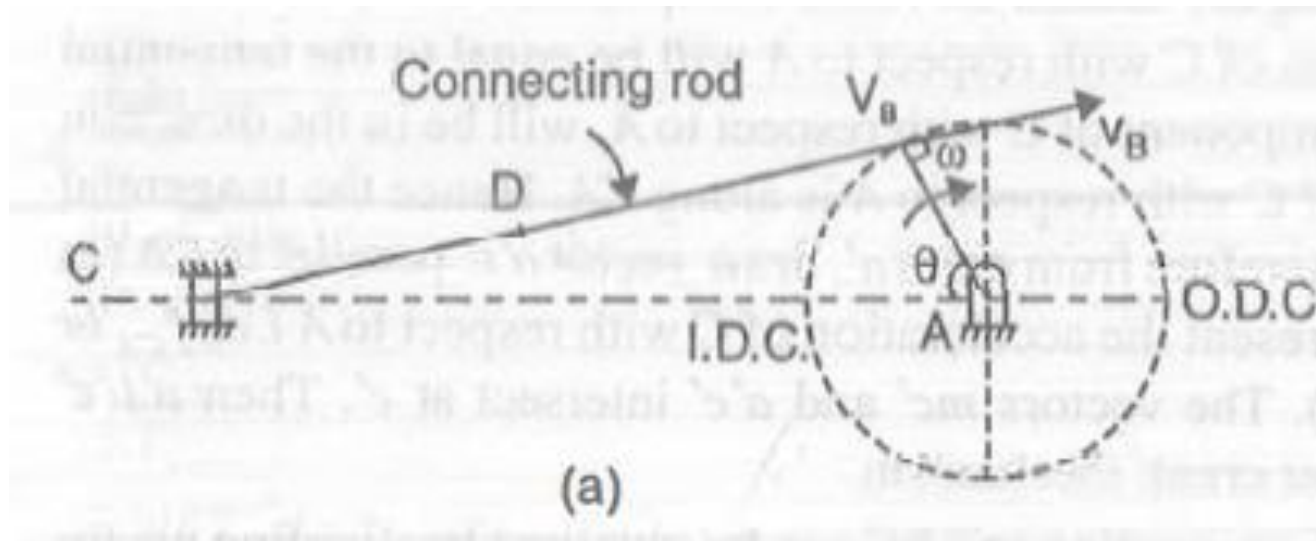
$$f^t_{BA} = \alpha \times BA$$

The total acceleration of **B** with respect to **A** is the vector sum of the above two components. This is represented by f_{BA} . Hence total acceleration of B with respect to A is given by-

$$f_{BA} = \text{vector sum of } f_{BA}^r + f_{BA}^t$$
$$= \text{vector sum of } \frac{v_{BA}^2}{BA} + \alpha \cdot BA$$

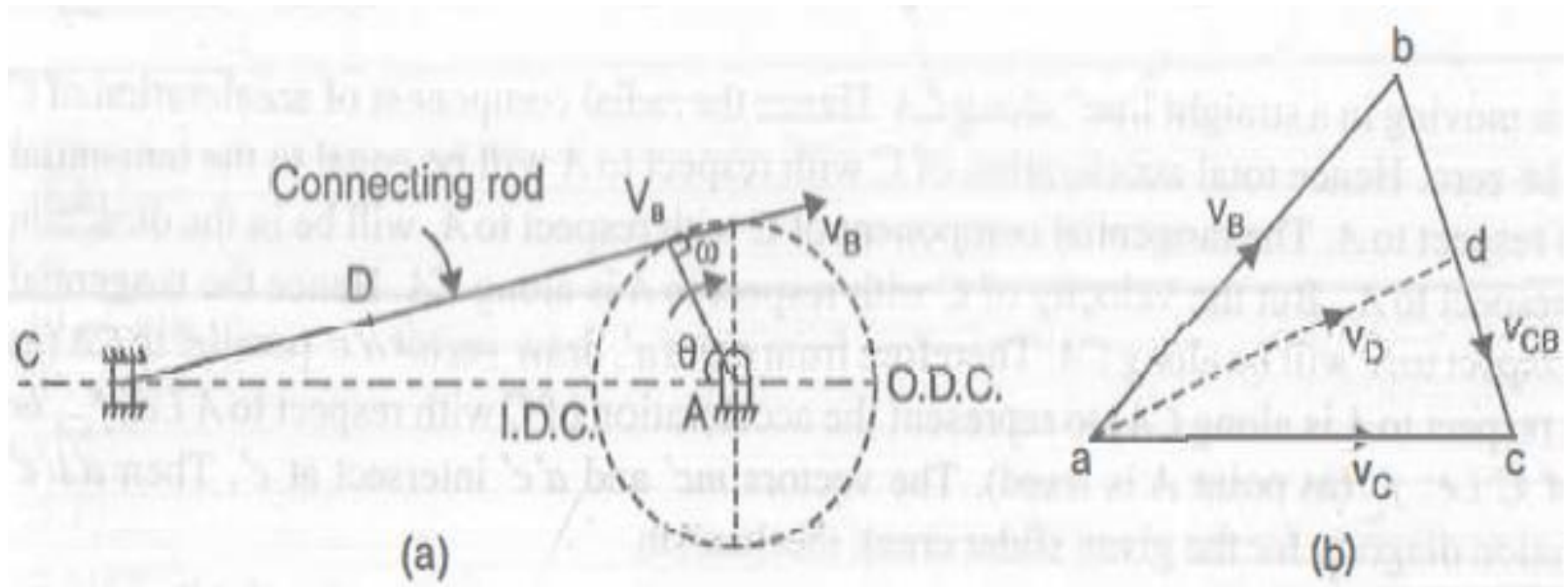
Acceleration Diagram for a Slider Crank Mechanism:

The Fig. (a) shows a slider crank mechanism in which the crank AB is rotating clockwise with a uniform angular velocity ω about fixed point A. Let the crank makes an angle θ with the inner dead centre (I.D.C.). The connecting rod is represented by CB.



Now the velocity of B with respect to A (or only the velocity of B as point A is fixed) is given by: V_{BA} or $V_B = \omega \times AB$

This velocity will be perpendicular to AB as shown in Fig. (a). Now the velocity diagram can be drawn as shown in Fig (b).



For drawing the acceleration diagram, first calculate the radial components of acceleration of different links.

Now radial acceleration of B with respect to A (or simply radial acceleration of B as point A is fixed) is given by-

$$f_{BA}^r \text{ or } f_B^r = \frac{v_{BA}^2 \text{ or } v_B^2}{AB} = \frac{v_B^2}{AB}$$

- As the point B is rotating with uniform angular velocity with respect to A, hence angular acceleration of B is zero. Therefore the tangential acceleration of B (which is equal to angular acceleration $\times$ BA) with respect to A, will be zero. Hence total acceleration of B with respect to A

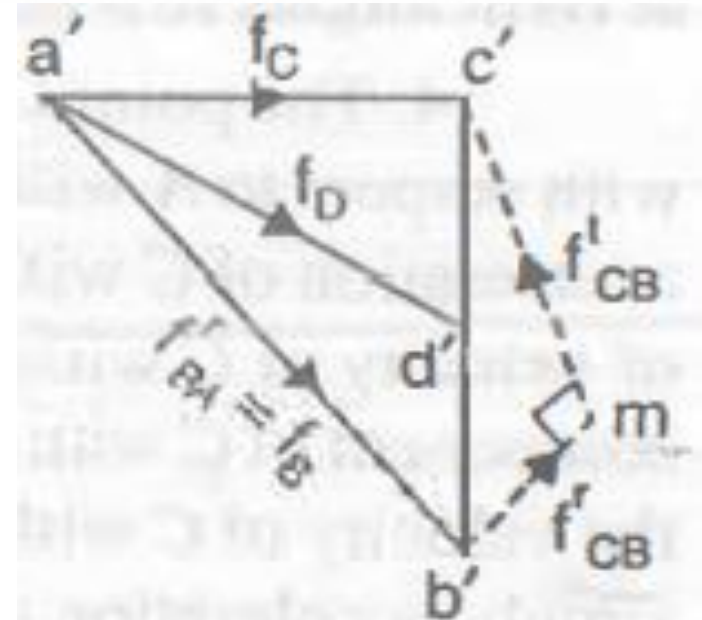
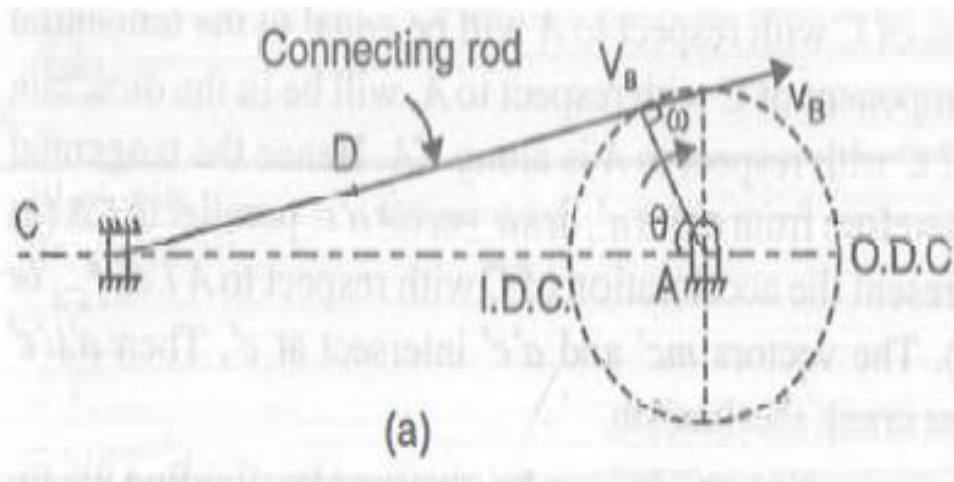
$$\begin{aligned} f_{BA} \text{ or } f_B &= f_{BA}^r \\ &= \frac{v_{BA}^2}{AB} \text{ or } \frac{v_B^2}{AB} \\ f_B &= \frac{v_B^2}{AB} \end{aligned}$$

- This acceleration is acting along BA. Hence the magnitude and direction of total acceleration of B with respect to A (simply total acceleration of B) is known.
- The radial component of acceleration of C with respect to B is given by-

$$f^r_{CB} = \frac{v^2_{CB}}{BC}.$$

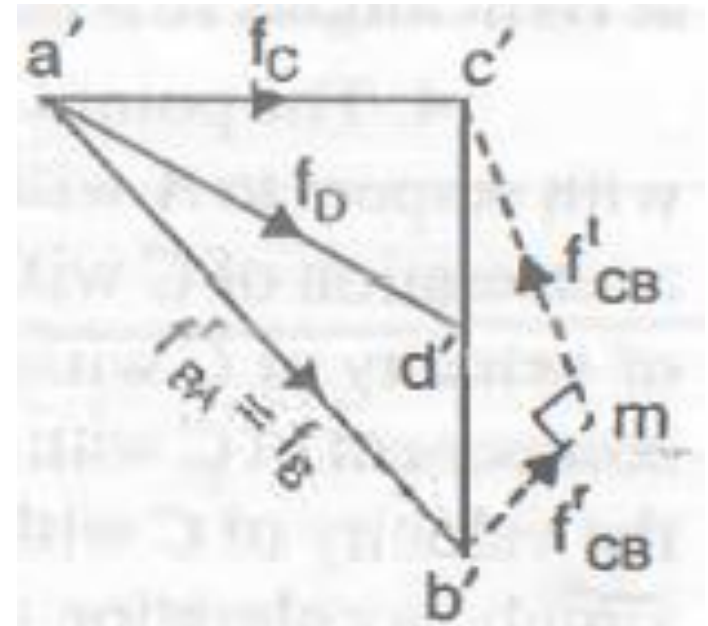
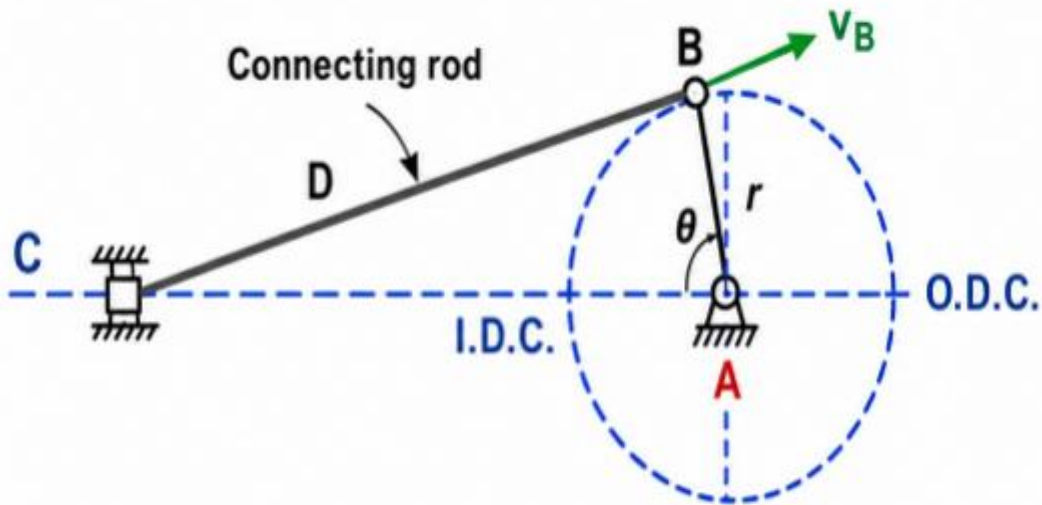
Acceleration Diagram

1. Choose a suitable scale for drawing the acceleration diagram. Take any point a' corresponding to fixed point A . Total acceleration of point B with respect to A is known. It is equal to $\frac{v_B^2}{AB}$ and is acting along BA . From a' draw vector $a'b'$ parallel to BA to represent the radial component of acceleration of B with respect to A (which is in this case is equal to f_B i.e., total acceleration of B with respect to A) such that vector $a'b' = f_B$.
2. The point C is having radial component of acceleration with respect to B and also tangential component of acceleration with respect to B . Radial component is acting along CB whereas tangential component is acting perpendicular to BC . Hence from b' , draw vector $b'm$ parallel to CB to represent the radial component of acceleration of C with respect to B (i.e., f'_{CB}). As the magnitude of f'_{CB} is known, therefore vector $b'm = f'_{CB} = \frac{v_{CB}^2}{BC}$.

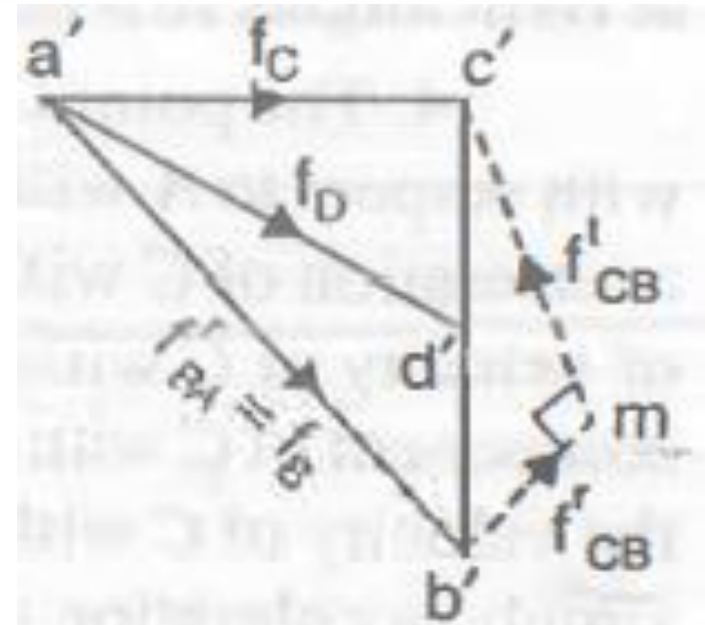
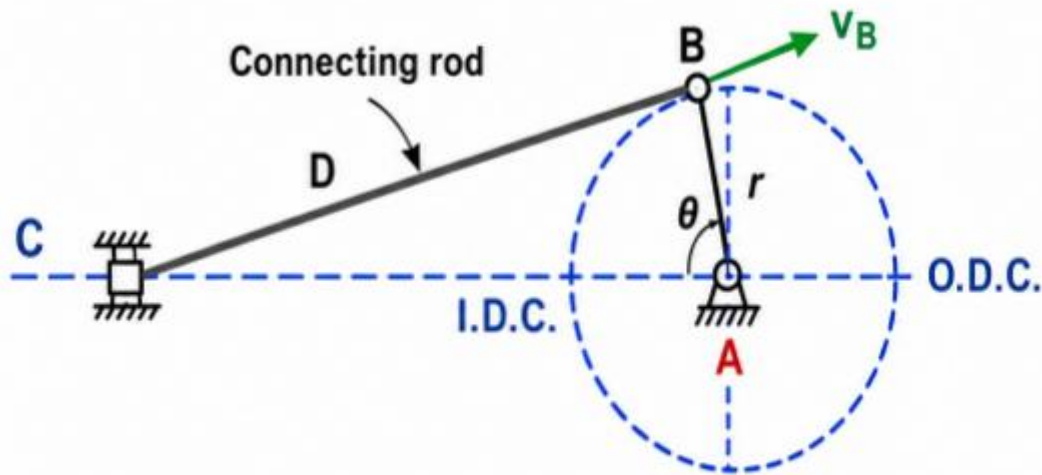


3. From point m , draw vector mc' perpendicular to BC to represent the tangential component of the acceleration of C with respect to B (i.e. f_{CB}^t). The magnitude f_{CB}^t is not yet known.

The line mc' will also be at right angles to $b'm$.



4. The point C is moving in a straight line* along CA . Hence the radial component of acceleration of C with respect to A will be zero. Hence total acceleration of C with respect to A will be equal to the tangential acceleration of C with respect to A . The tangential component of C with respect to A , will be in the direction of velocity of C with respect to A . But the velocity of C with respect to A is along CA . Hence the tangential component of C with respect to A will be along CA . Therefore from point a' , draw vector $a'c'$ parallel to CA (as the velocity of C with respect to A is along CA) to represent the acceleration of C with respect to A i.e., f_{CA} or simply acceleration of C i.e., f_C (as point A is fixed). The vectors mc' and $a'c'$ intersect at c' . Then $a'b'c'$ represents the acceleration diagram for the given slider crank



5. The total acceleration of any point D on the connecting rod BC can be obtained by dividing vector $b'c'$ at d' in the same ratio as D divides BC . This means that

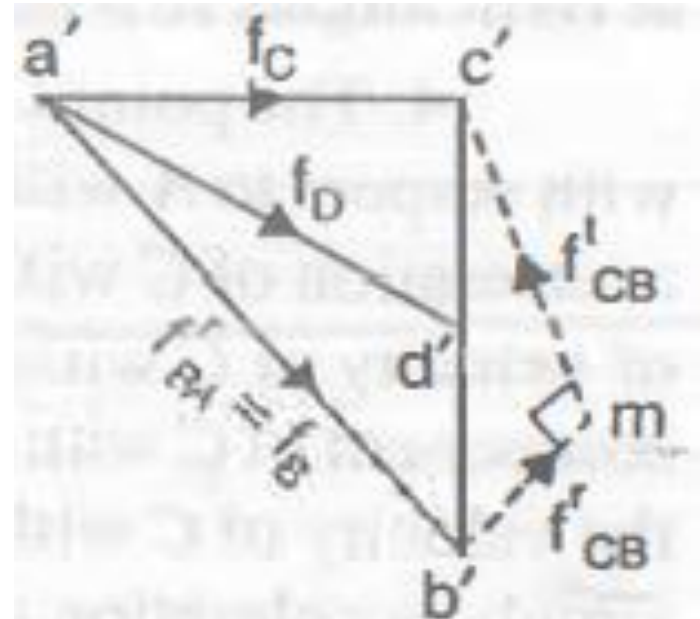
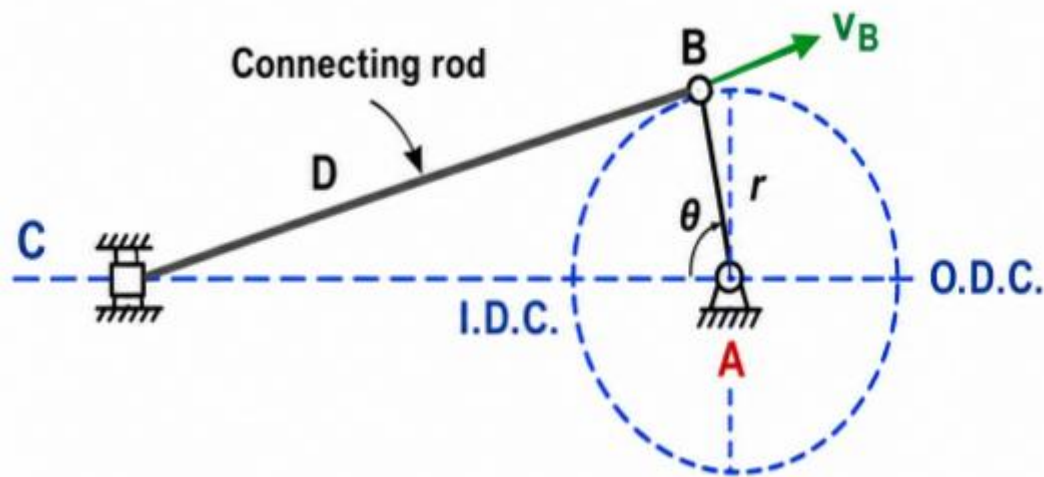
$$\frac{b'd'}{b'c'} = \frac{BD}{BC},$$

$$\therefore b'd' = \frac{BD}{BC} \times b'c'.$$

6. The angular acceleration of connecting rod CB is given by,

$$f'_{CB} = \alpha_{CB} \times CB$$

$$\therefore \alpha_{CB} = \frac{f'_{CB}}{CB} \quad \text{where } \alpha_{CB} = \text{Angular acceleration of } CB.$$



Q 1: The crank of a slider crank mechanism is 15 cm and the connecting rod is 60 cm long. The crank makes 300 r.p.m. in the clockwise direction. When it has turned 45° from the inner dead centre position, determine :

- (i) Acceleration of the mid-point of the connecting rod and*
- (ii) Angular acceleration of the connecting rod*

Crank length = $AB = 15$ cm, Connecting rod length = $BC = 60$ cm

Crank speed = $N = 300$ r.p.m., Angle turned by crank from I.D.C. = 45°

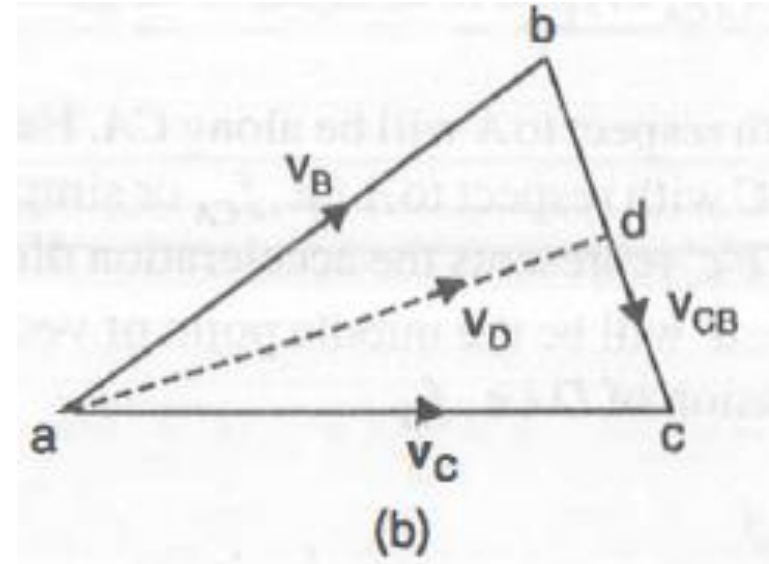
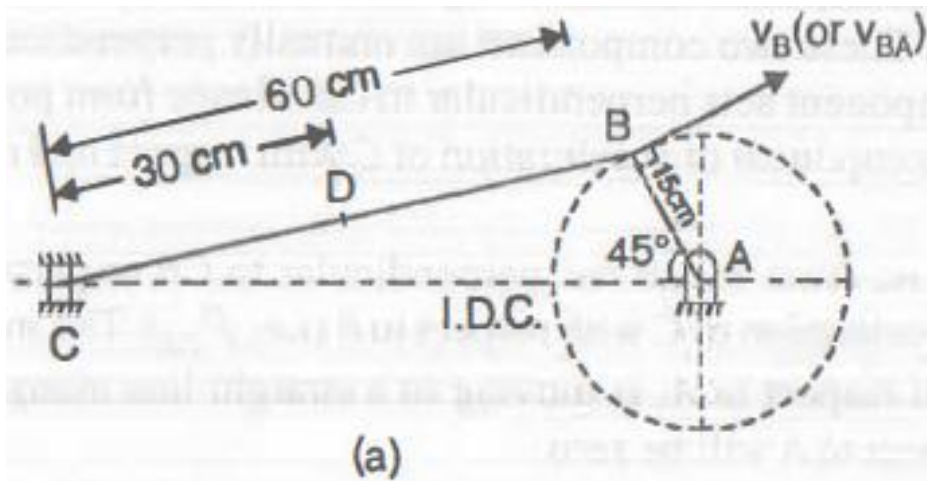
Angular velocity of crank, $\omega = \frac{2\pi N}{60} = \frac{2\pi \times 300}{60} = 31.42 \text{ rad/s}$

$\therefore$ Velocity of B with respect to A (or velocity of B only as point A is fixed) is given by

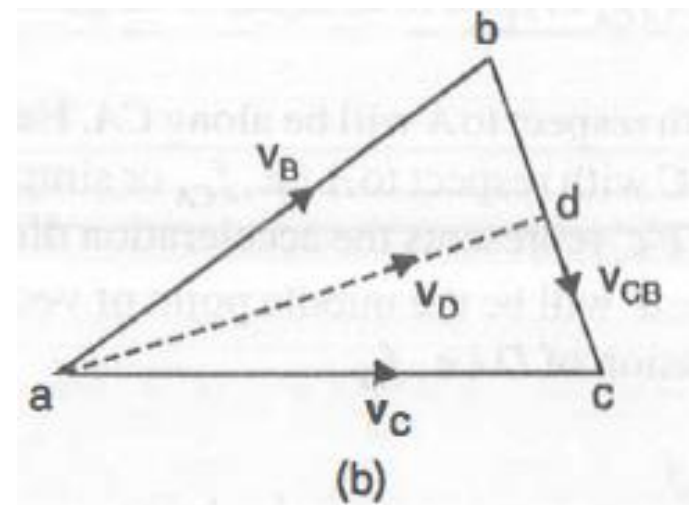
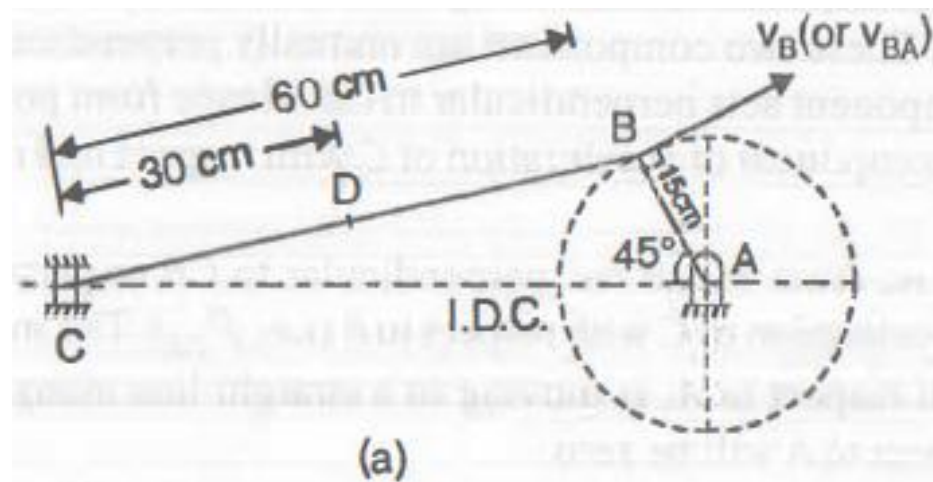
$$v_{BA} = v_B = \omega \times AB = 31.42 \times 15 = 471.3 \text{ cm/s}$$

The direction of v_B will be perpendicular to AB .

I . Take any point a (fixed point). Choose a suitable velocity scale and draw vector ab perpendicular to AB to represent the velocity of B with respect to A or simply the velocity of B such that vector $ab = V_{BA}$ or $V_B = 471.3 \text{ cm/s}$.



2. As the point C is moving with respect to B and also with respect to A . The velocity of C with respect to B is perpendicular to BC whereas the velocity of C with respect to A is along CA (i.e., horizontally). Hence from point b , draw vector be perpendicular to BC to represent the velocity of C with respect to B and from point a draw vector ac horizontally to represent the velocity of C with respect to A . The vectors be and bc intersect at point ac . Then abc is the velocity diagram



3. As the point D is the middle point of CB , hence corresponding point d will be the middle point of vector bc . Join ad . Then vector ad represents the velocity of point D .

- From velocity diagram
- (i) Velocity of D, $v_D = \text{vector } ad = 410 \text{ cm/s}$
- (ii) Velocity of C, $v_c = \text{vector } ac = 400 \text{ cm/s}$
- (iii) Velocity of C with respect to B, $v_{CB} = \text{vector } bc = 340 \text{ cm/s}$.
- **Acceleration diagram:**

Before drawing acceleration diagram, we must calculate the radial acceleration $\left(i.e., \frac{v^2}{r} \right)$ of each link.

$\therefore$ Radial acceleration of B with respect to A (or simply radial acceleration of B as point A is fixed) is given by,

$$f_{BA}^r \text{ or } f_B^r = \frac{v_B^2}{AB} = \frac{471.3^2}{15} = 14808 \text{ cm/s}^2$$

Radial acceleration of C with respect to B is given by,

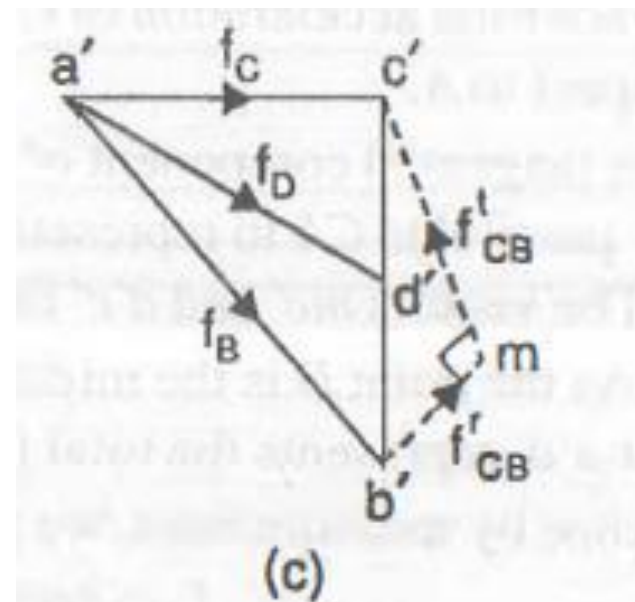
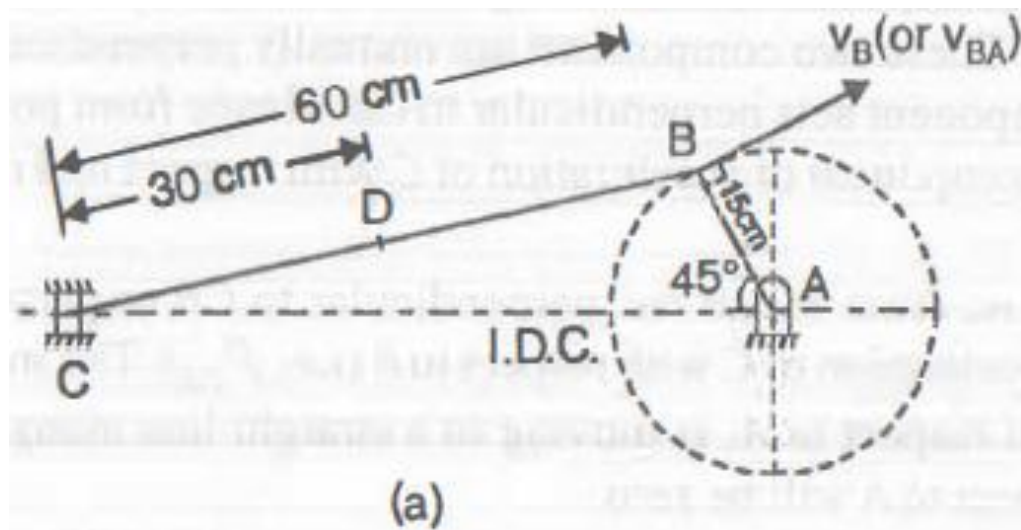
$$f_{CB}^r = \frac{v_{CB}^2}{CB} = \frac{340^2}{60} \quad (\because v_{CB} = 340 \text{ and } CB = 60 \text{ cm})$$

$$= 1926.7 \text{ cm/s}^2$$

As the point B is rotating with uniform angular velocity with respect to A , hence angular acceleration of B with respect to A is zero. Therefore the tangential acceleration of B with respect to A will be zero. Hence total acceleration of B with respect to A will be equal to radial acceleration of B with respect to A and will be acting along BA .

$$f_{BA} \quad \text{or} \quad f_B = f'_{BA} \quad \text{or} \quad f'_B = 14808 \text{ cm/s.}$$

1. Choose a suitable scale for drawing acceleration diagram. Take any point a' corresponding to fixed point A . Total acceleration of B with respect to A is known. It is equal to radial acceleration of B and is acting along BA . From a' draw vector $a'b'$ parallel to BA to represent the radial acceleration of B with respect to A (which is in this case is equal to f_{BA} or f_B i.e., total acceleration of B). Hence cut vector $a'b' = f_B$ or $f'_{BA} = 14808$



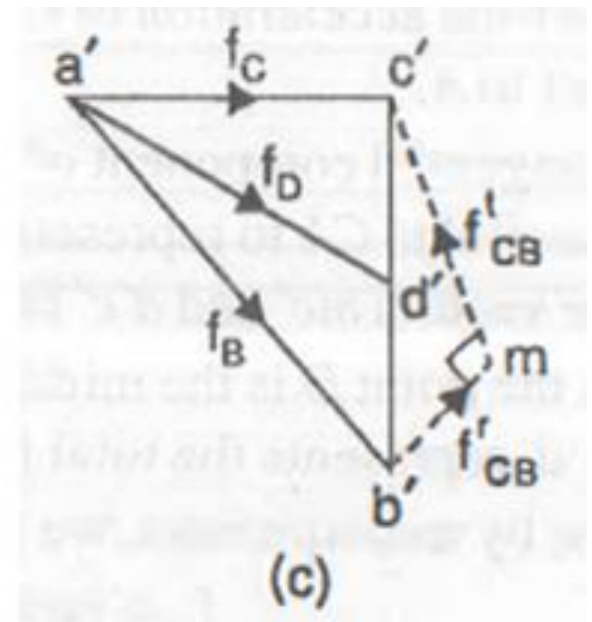
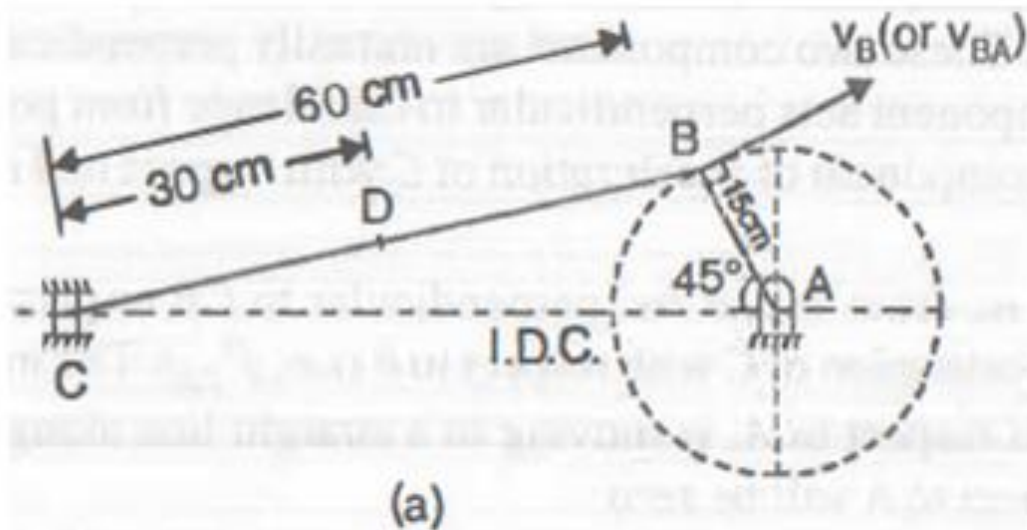
$$f_{CB}^r = 1926.7 \text{ cm/s}^2.$$


3. Now from point m , draw vector mc' perpendicular to CB or perpendicular to $b'm$ to represent the tangential component of acceleration of C with respect to B (i.e., f'_{CB}). The magnitude to f'_{CB} is not yet known.

4. The point C with respect to A , is moving in a straight line along CA . Hence radial component of acceleration of C with respect to A will be zero.

Hence total acceleration of C with respect to A (i.e., f_{CA} or f_C) will be equal to tangential acceleration of C with respect to A .

The tangential component of acceleration of C with respect to A will be along CA . Hence from a' , draw vector $a'c'$ parallel to CA to represent the acceleration of C with respect to A i.e., f_{CA} or simply acceleration of C i.e., f_C . The vectors mc' and $a'c'$ intersect at c' . Then $a'b'c'$ represents the acceleration diagram.



5. As the point D is the middle point of CB , hence d' will be the middle point of vector $b'c'$. Join $a'd'$. The vector $a'd'$ represents the total (or absolute) acceleration of D i.e., f_D .

Hence by measurement, we get

$$f_D = \text{vector } a'd' = 11700 \text{ cm/s}^2$$

(i) $\therefore$ Acceleration of mid-point of connecting rod (CB) is $f_D = 11700 \text{ cm/s}^2$. Ans.

(ii) Angular acceleration of connecting rod (CB). We know that

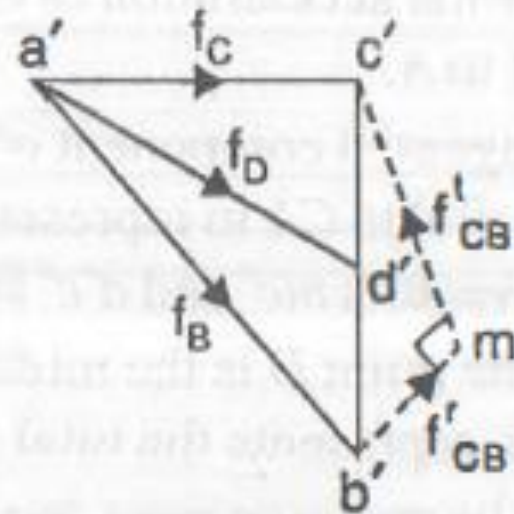
$$f'_{CB} = \alpha_{CB} \times CB$$

where f'_{CB} = Tangential acceleration of C with respect to B (or tangential acceleration of connecting rod CB).
 = Vector $mc' = 10300 \text{ cm/s}^2$ (by measurement)

α_{CB} = Angular acceleration of connecting rod

$$= \frac{f'_{CB}}{\text{Length } CB}$$

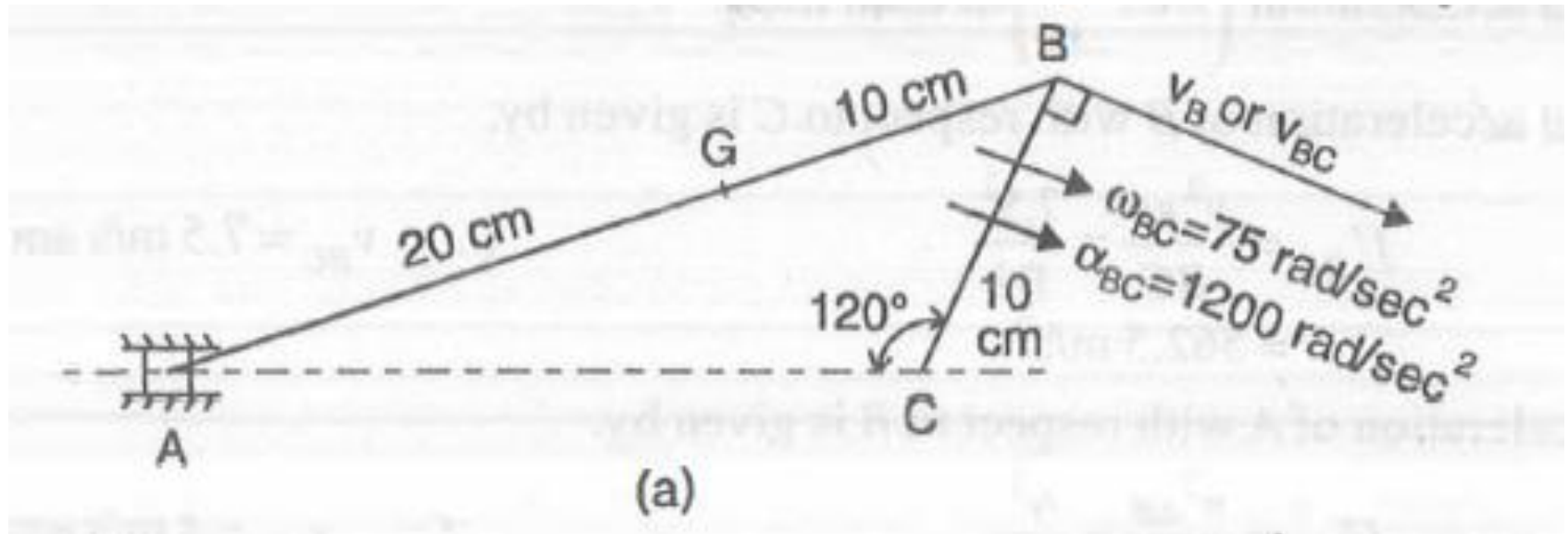
$$= \frac{10300}{60} = 171.67 \text{ rad/s}^2. \text{ Ans.}$$



(c)

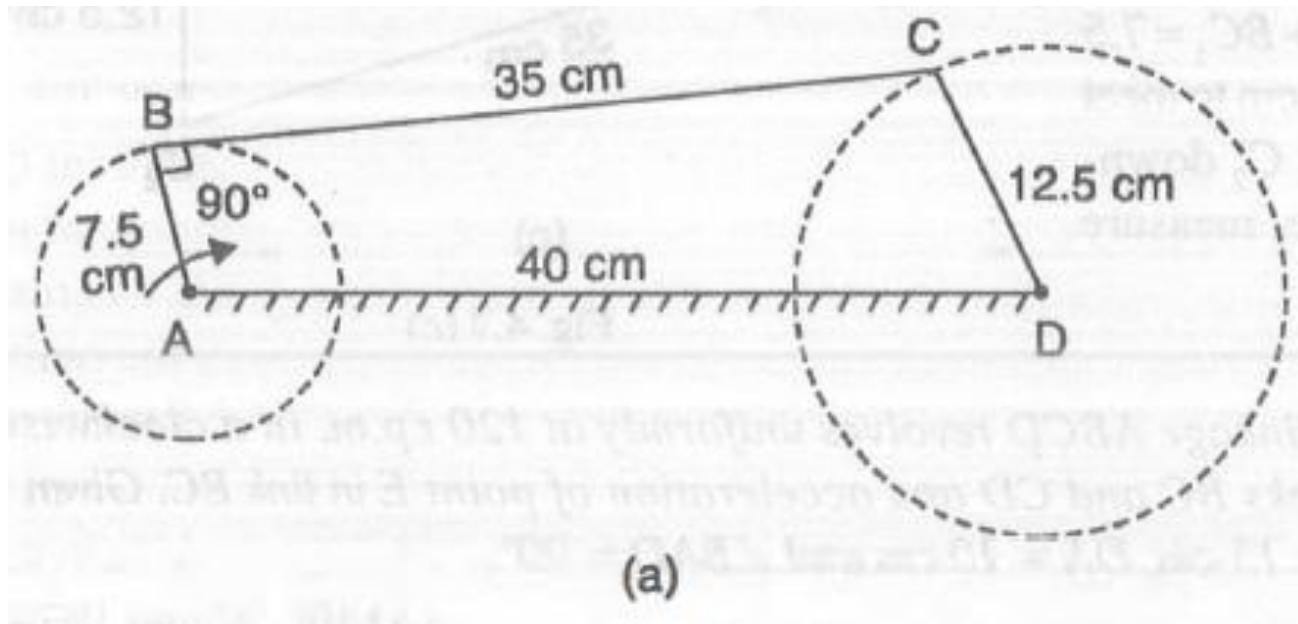
Home Work:

- A reciprocating engine mechanism is shown in Fig. (a). The crank $CB=10$ cm and connecting rod $BA=30$ cm with the centre of gravity G , 10 cm from B . In the position shown, the crank has a velocity of 75 rad/s and an angular acceleration of 1200 rad/s². Find : (a) the velocity and acceleration of G and (b) the angular velocity and angular acceleration of AB .



Q 2: A four bar kinematic chain is represented by a quadrilateral ABCD in which AD is fixed and is 400 mm long. The crank AB 75 mm long rotates in a clockwise direction at 120 r.p.m. and drives the link CD 125 mm long by means of the connecting link BC 350 mm long. Determine the angle through which CD oscillates and find the angular velocities of the links BC and CD in one of the positions when BC is perpendicular to AB.

FBD:



- *AD is fixed. Length $AD=400\text{ mm}=40\text{ cm}$, $AB=75\text{ mm}=7.5\text{ cm}$,*
- *$N=120\text{ r.p.m.}$, $CD=125\text{ mm}=12.5\text{ cm}$, $BC=350\text{ mm}=35\text{ cm}$*

Now $N_{BA} = 120\text{ r.p.m.}$

$$\therefore \omega_{BA} = \frac{2\pi \times 120}{60} = 12.57\text{ rad/s}$$

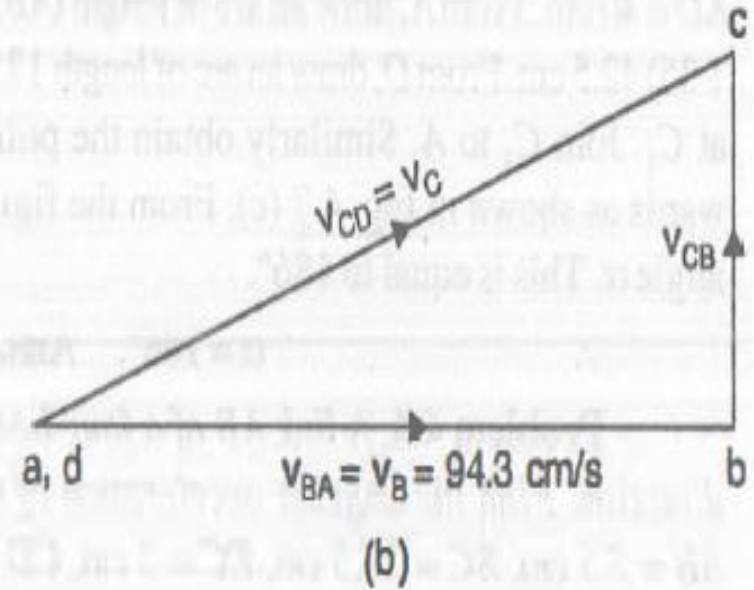
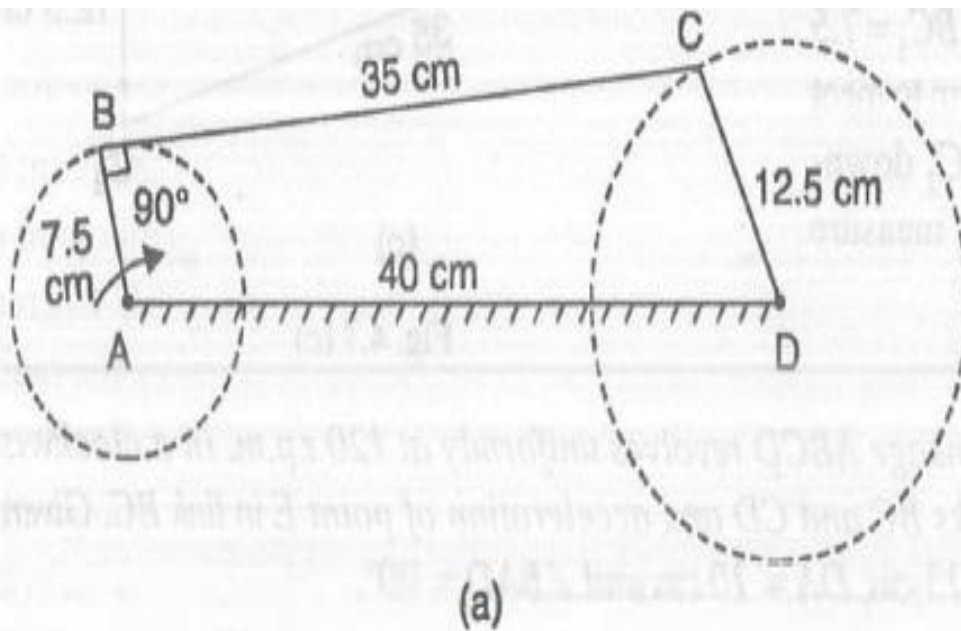
Now velocity of B with respect to A ,

$$\begin{aligned} v_{BA} &= \omega_{BA} \times \text{Length } BA \\ &= 12.57 \times 7.5 = 94.3\text{ cm/s} \end{aligned}$$

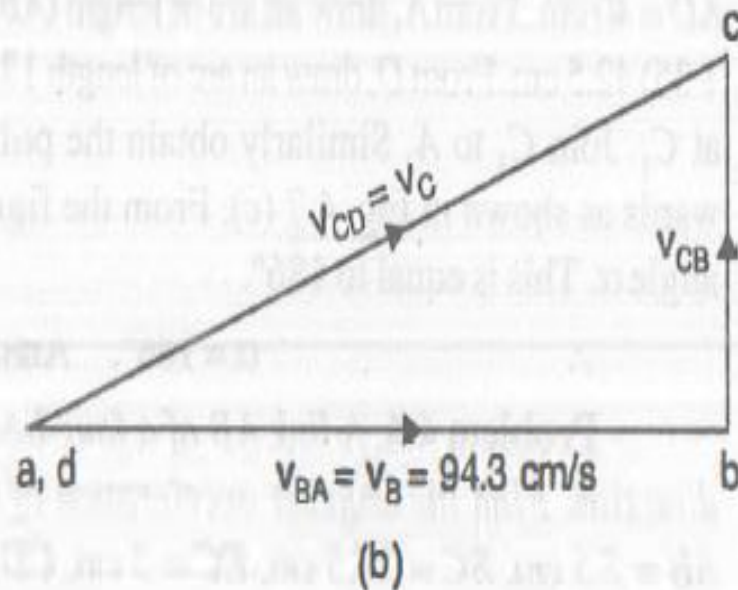
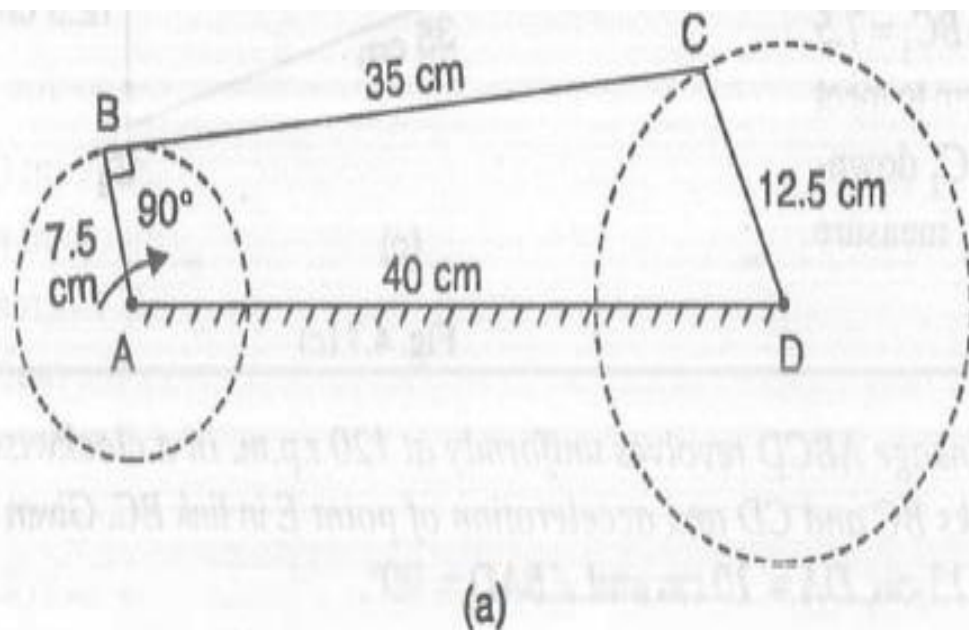
As point A is fixed, hence velocity of B (or velocity of B with respect to A) will be 94.3 cm/s .

$$\therefore v_{BA} = v_B = 94.3\text{ cm/s.}$$

1 . Since A and D are fixed points, therefore these points (a, d) lie at one place in velocity diagram. Choose a suitable velocity scale and from point ' a ' draw vector ab perpendicular to AB to represent the velocity of B with respect to A (i.e., V_{BA}). Take $ab=94.3$ cm/s.



2. The point C moves with respect to point B and also with respect to point D . The velocity of C with respect to B is perpendicular to BC whereas the velocity of C with respect to D is perpendicular to DC . Hence from point ' b ', draw vector be perpendicular to BC . Since link AD is fixed hence point ' a ' and ' d ' will coincide in the velocity diagram. From point $(d \text{ or } a)$, draw vector dc (or ac) perpendicular to DC so as to intersect vector be in ' c '. Then abc is the velocity diagram :



Vector bc = Velocity of C with respect to B (i.e., v_{CB})

Vector dc = Velocity of C with respect to D (i.e., v_{CD}). As point D is fixed, this is also the velocity of point C (i.e., v_C).

By measurement, from velocity diagram, we get

$$v_{CB} = \text{vector } bc = 36.75 \text{ cm/s}$$

and v_{CD} or $v_C = \text{Vector } dc = 100.8 \text{ cm/s}$

(i) Angular velocities of the links BC and CD

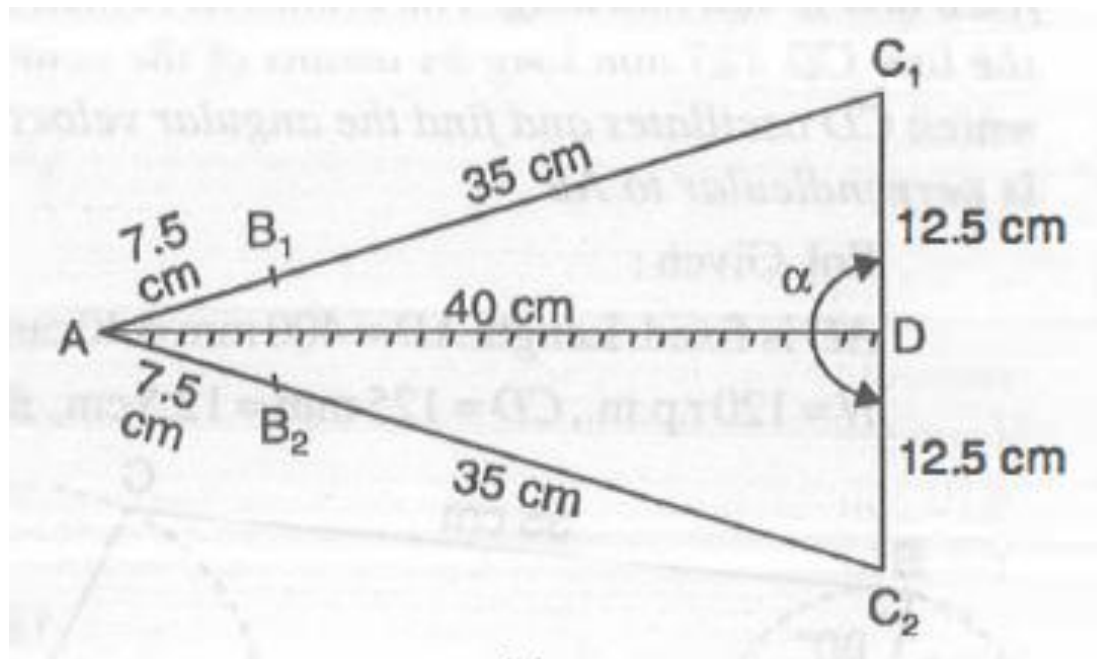
Let ω_{CB} = Angular velocity of link CB

ω_{CD} = Angular velocity of link CD

Then $\omega_{CB} = \frac{v_{CB}}{CB} = \frac{36.75}{35} = 1.05 \text{ rad/s. Ans.}$

and $\omega_{CD} = \frac{v_{CD}}{CD} = \frac{100.8}{125} = 8.06 \text{ rad/s. Ans.}$

- **(ii) Angle through which CD oscillates:**
- Let α = angle through which CD oscillates. Take $AD = 40$ cm. From A, draw an arc of length $(AB_1 + B_1C_1 = 7.5 + 35) 42.5$ cm.
- From D, draw an arc of length 12.5 cm to meet at C_1 .
- Join C_1 to A. Similarly obtain the point C_2 downwards as shown in Fig. From the figure, measure angle α . This is equal to 186° .



Home work:

- A link AB of a four-bar linkage ABCD revolves uniformly at 120 r.p.m. in a clockwise direction. Find the angular acceleration of links BC and CD and acceleration of point E in link BC. Given :
- $AB=7.5\text{cm}$, $BC=17.5\text{cm}$, $EC=5\text{cm}$, $CD=15\text{cm}$, $DA=10\text{cm}$ and Angle $BAD=90^\circ$.

